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ROOFTOP TENT ANNEX SYSTEM

Abstract

A tent system includes a base configured to mount to a vehicle, a base canopy, a frame system coupled to the base and configured to support the base canopy, and an annex. The annex includes an annex canopy and an annex frame assembly configured to support the annex canopy. The annex is removably coupled to the base and the base canopy.

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Background/Summary

BACKGROUND

Field

[0001] The present disclosure relates to tent apparatuses, systems, and methods, for example, adaptable rooftop tent apparatuses, systems, and methods having annex portions that can expand the interior volume of the rooftop tent apparatus.

Background

[0002] Current rooftop tents can utilize a plurality of tent poles attached to a base to support a main tent canopy and create an internal volume for a user on a vehicle rooftop. However, rooftop tents provide little additional space useful for camping. For example, the general shape of a rooftop of tent provides space to sit in or lie down, but may be limiting for standing room, for example, when changing clothes. Further, rooftop tents are generally limited with storage space and provide limited shelter to the neighboring campsite area from the elements, such as sun, rain, or snow.

BRIEF SUMMARY

[0003] Accordingly, there is a need to, for example, provide an annex that can be removably coupled to a rooftop tent to provide additional storage and functional space such that campers have a place to change clothes, sit or stand to shelter from the sun or inclement weather conditions, or provide additional storage for equipment while camping.

[0004] In some embodiments, a tent system includes a base, a base canopy, a frame system, and an annex. In some embodiments, the base is configured to mount to a vehicle. In some embodiments, the frame system is coupled to the base and configured to support the canopy. In some embodiments, the annex includes an annex canopy and an annex frame assembly configured to support the annex canopy. In some embodiments, the annex is removably coupled to the base and the base canopy.

[0005] In some embodiments, the annex canopy is removably coupled to the base canopy and the annex frame assembly is removably coupled to the base. In some embodiments, the annex canopy is removably coupled to the base canopy with at least one zipper. In some embodiments, the annex frame assembly is coupled to a bottom side of the base. In some embodiments, the base canopy includes a door disposed on a first surface, and the annex canopy surrounds the door.

[0006] In some embodiments, the tent system further includes a ladder coupled to the base and extending between a floor and the base. In some embodiments, the annex completely surrounds the ladder.

[0007] In some embodiments, the annex canopy includes a lower opening portion. In some embodiments, the lower opening portion includes a plurality of side sections. In some embodiments, at least two of the plurality of side sections are configured to be in one of an opened configuration and a closed configuration.

[0008] In some embodiments, a tent system includes a base, a base canopy, a frame system, and an annex. In some embodiments, the base is configured to mount to a vehicle. In some embodiments, the frame system is coupled to the base and configured to support the base canopy. In some embodiments, the annex includes an annex canopy and an annex frame assembly configured to support the annex canopy. In some embodiments, the annex frame assembly is removably coupled to a bottom side of the base.

[0009] In some embodiments, the annex frame assembly is removably coupled to a base bracket disposed on the bottom side of the base. In some embodiments, the annex frame assembly includes a fastening member configured to slideably engage with an interior space formed in the base bracket. In some embodiments, the fastening member includes a T-bolt slideably disposed in the interior space and an actuator is threadably coupled with the T-bolt and configured to secure the annex frame assembly to the base bracket.

[0010] In some embodiments, the base bracket includes a first flange and a second flange spaced apart from the first flange to form a slot between the flanges. In some embodiments, the first and second flanges include engagement surfaces that have a shape corresponding to a connection portion of the annex frame assembly. In some embodiments, the engagement surfaces are configured to engage the connection portion of the annex frame assembly. In some embodiments, the connection portion of the annex frame assembly is cylindrical and the engagement surfaces of the first and second flanges are concave.

[0011] In some embodiments, the base bracket includes a stopper flange disposed in the interior space and configured to block movement of the fastening member. In some embodiments, the annex frame assembly does not extend through the base canopy.

[0012] In some embodiments, a tent system includes a base, a base canopy, a frame system, and an annex. In some embodiments, the base is configured to mount to a vehicle. In some embodiments, the frame system is coupled to the base and configured to support the base canopy. In some embodiments, the annex is configured to extend an interior space of the tent system in a lateral direction away from the base canopy and in a vertical direction toward the ground. In some embodiments, the annex includes an annex canopy and an annex frame assembly configured to support the annex canopy. In some embodiments, the annex canopy includes a height extendable portion configured to adjust a height of the annex canopy relative to the ground.

[0013] In some embodiments, the height extendable portion includes a first extendable section having a first section height, an upper coupling member disposed on an upper edge of the first extendable section, and a first lower coupling member disposed on a lower edge of the first extendable section. In some embodiments, the annex canopy is configured to be disposed in a first shortened configuration such that the first lower coupling member is coupled to the upper coupling member and the height of the annex canopy is reduced by the first section height.

[0014] In some embodiments, the upper coupling member and the lower coupling member include a zipper, a toggle and loop, a button and hole, or a hook-and-loop fastener. In some embodiments, the upper coupling member extends along a majority of the upper edge of the first extendable section, and the first lower coupling member extends along a majority of the lower edge of the first extendable section.

[0015] In some embodiments, the height extendable portion further includes a second extendable section having a second section height and extending from the lower edge of the first extendable section. In some embodiments, the first lower coupling member is disposed on an upper edge of the second extendable section, and a second lower coupling member is disposed on a lower edge of the second extendable section.

[0016] In some embodiments, the annex canopy is configured to be disposed in a second shortened configuration such that the second lower coupling member is coupled to the upper coupling member and the height of the annex canopy is reduced by the first and second section heights. In some embodiments, the annex canopy is configured to be disposed in a second shortened configuration such that the second lower coupling member is coupled to the first lower coupling member and the height of the annex canopy is reduced by the second section height.

[0017] In some embodiments, the first section height is equal to the second section height. In some embodiments, the first section height is greater than the second section height.

[0018] In some embodiments, a tent annex system includes an annex canopy and an annex frame assembly. In some embodiments, the annex frame assembly includes a fastening member and a support pole. In some embodiments, the support pole is configured to support the annex canopy. In

some embodiments, the fastening member of the annex frame is configured to removably couple to a base of a tent system. In some embodiments, the fastening member of the annex frame is configured to removably couple to a bottom side of a base of a tent system. In some embodiments, the annex canopy is configured to removably couple to a tent canopy of the tent system. In some embodiments, the annex system is configured to extend an interior space of the tent system in a lateral direction away from the vehicle and a vertical direction toward a floor.
[0019] Implementations of any of the techniques described above may include a system, a method, a process, a device, and/or an apparatus. The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features will be apparent from the description and drawings, and from the claims.
[0020] Further features and advantages of the disclosure, as well as the structure and operation of various embodiments of the disclosure, are described in detail below with reference to the accompanying drawings. It is noted that the disclosure is not limited to the specific embodiments described herein. Such embodiments are presented herein for illustrative purposes only. Additional embodiments will be apparent to persons skilled in the relevant art(s) based on the teachings contained herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0021] The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate the embodiments and, together with the description, further serve to explain the principles and to enable a person skilled in the relevant art(s) to make and use the embodiments. Objects and advantages of illustrative, non-limiting embodiments will become more apparent by describing them in detail with reference to the attached drawings.
[0022] FIG. **1** illustrates an exploded perspective view of an annex and tent, according to an embodiment.
[0023] FIG. **2** illustrates a perspective view of an annex coupled with the tent, according to an embodiment.
[0024] FIG. **3** illustrates a side view of an annex and tent mounted on a vehicle, according to an embodiment.
[0025] FIG. **4** illustrates a bottom perspective view of an annex frame assembly coupled with a tent frame system, according to an embodiment.
[0026] FIG. **5** illustrates a detailed perspective view of an annex frame assembly coupled with a tent frame assembly, according to an embodiment.
[0027] FIG. **6** illustrates a cross-sectional view of the annex frame assembly along sectional line **6-6** shown in FIG. **5**, according to an embodiment.
[0028] FIG. **7** illustrates a cross-sectional view of the annex frame assembly along sectional line **7-7** shown in FIG. **5**, according to an embodiment.
[0029] FIG. **8** illustrates a side view of an annex and tent, with the annex in an extended configuration, according to an embodiment.
[0030] FIG. **9** illustrates a side view of an annex and tent, with the annex in a shortened configuration, according to an embodiment.
[0031] FIG. **10** illustrates an interior perspective view of an annex showing first, second, and third extendable sections, according to an embodiment.
[0032] FIG. **11** illustrates an interior perspective view of an annex in an intermediate shortened configuration, according to an embodiment.
[0033] FIG. **12** illustrates an exterior perspective view of an annex in the intermediate shortened configuration, according to an embodiment.

[0034] FIG. **13** illustrates a perspective view of a tent with an annex flap, according to an embodiment.
[0035] FIG. **14**A illustrates a perspective view of a tent with a zipper configuration for coupling with the annex, according to an embodiment.
[0036] FIGS. **14**B and **14**C illustrate detailed views of the zipper configuration shown in FIG. **14**, according to an embodiment.
[0037] FIG. **15** illustrates a perspective view of a tent with a zipper configuration for coupling with the annex, according to an embodiment.
[0038] FIG. **16** illustrates a top view of an annex floor, according to an embodiment.
[0039] FIG. **17** illustrates an attachment clip for coupling an annex to a base of a tent frame system, according to an embodiment.
[0040] The features and advantages of the embodiments will become more apparent from the detailed description set forth below when taken in conjunction with the drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

DETAILED DESCRIPTION

[0041] Embodiments of the present disclosure are described in detail with reference to embodiments thereof as illustrated in the accompanying drawings. References to “one embodiment,” “an embodiment,” “some embodiments,” etc., indicate that the embodiment(s) described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.
[0042] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “on,” “upper,” “opposite” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or in operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (e.g., rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
[0043] The term “about” or “substantially” or “approximately” as used herein indicates the value of a given quantity that can vary based on a particular technology. Based on the particular technology, the term “about” or “substantially” or “approximately” can indicate a value of a given quantity that varies within, for example, 1-15% of the value (e.g., ±1%, ±2%, ±5%, ±10%, or ±15% of the value).
[0044] Before describing such embodiments in more detail, it is instructive to present example environments in which embodiments of the present disclosure can be implemented.

Example Tent

[0045] Tent system **200** can be configured to expand and collapse a frame system (e.g., frame system **400** shown in FIGS. **1-4**) and a base canopy (e.g., canopy system **500** shown in FIGS. **1-4**) coupled to the frame system. As shown in FIGS. **1-4**, tent system **200** can include base system **300**, frame system **400**, canopy system **500**, and/or rain fly system **600** (not shown). Base system **300** can be configured to mount to a vehicle and support frame system **400**, canopy system **500**, and/or rain fly system **600**. Base system **300** can be coupled to frame system **400** and canopy system **500**. Frame system **400** can be configured to support canopy system **500**. Frame system **400** can be coupled to base system **300** and canopy system **500**. Further details and implementations of the tent system **200** and associated components are described in U.S. Patent Application Ser. No. 63/300,502, filed on Jan. 18, 2022, the entire contents of which is incorporated by reference herein.

[0046] Base system **300** can be configured to mount to a vehicle (e.g., vehicle **342** shown in FIG. **3**). As shown in FIGS. **1-4**, base system **300** can include first base member **312** and second base member **314**. First base member **312** can be configured to form a platform for tent system **200** atop a vehicle. First base member **312** can be further configured to receive a ladder. Second base member **314** can be configured to form a platform for tent system **200** atop a vehicle. Second base member **314** can be further configured to attach (e.g., be fixed) to a rack of a vehicle. As shown in FIG. **4**, for example, first and second base members **312**, **314** can be coupled to each other, for example, via hinge **318**. Hinge **318** can be configured to open and close first and second base members **312**, **314** to expand and contract tent system **200**.
[0047] Frame system **400** can include first frame member **410**, second frame member **420**, third frame member **430**, and fourth frame member **440** as shown, for example, in FIG. **4**. In some embodiments, frame members **410**, **420**, **430**, **440** can include aluminum, titanium, fiberglass, a metal, a ceramic, a polymer, a combination thereof, and/or any other rigid material. In some embodiments, frame members **410**, **420**, **430**, **440** can be configured to expand canopy system **500** to form an internal volume for a user. For example, in open configuration **10**, frame members **410**, **420**, **430**, **440** can contact canopy system **500** when a canopy is attached.
[0048] First frame member **410** can be configured to operate as a first knee connection of frame system **400**. As shown in FIG. **4**, first frame member **410** can include first support **412**, first joint **414**, and first leg **416**. First support **412** can be configured to pivot about first connection **411** to first base member **312**. First support **412** can be coupled to first base member **312** of base system **300**. First support **412** can be coupled to first leg **416** via first joint **414**. First joint **414** can be configured to pivot first leg **416** about distal end of first support **412**, for example, similar to a human knee joint. First leg **416** can be configured to contact canopy system **500** (e.g., canopy sleeve **560** shown in FIGS. **1** and **2**) and to expand canopy system **500** into an expanded position. Similar elements and features of second frame member **420**, third frame member **430**, and fourth frame member **440** are described in U.S. Patent Application Ser. No. 63/300,502, filed on Jan. 18, 2022, the entire contents of which is incorporated by reference herein.
[0049] Canopy system **500** can be configured to receive frame system **400** and provide weather protection (e.g., rain, sun, heat, wind, snow, cold, etc.). Canopy system **500** can be further configured to support rain fly system **600**. As shown in FIGS. **1-3**, in some embodiments, canopy system **500** can include canopy **510** and canopy sleeves **560**. Canopy **510** can be further configured to provide an internal volume for a user. In some embodiments, canopy **510** can include a polymer, a plastic, a thermoplastic, an elastomer, polyester (e.g., Mylar®), thermoplastic polyurethane (TPU), polyethylene (e.g., Dyneema®), ultra-high molecular weight polyethylene (UHMWPE) (e.g., Dyneema® composite fiber (DCF)), nylon, silicone, poly-cotton, canvas,), a combination thereof, and/or any other water resistant material.
[0050] As shown in FIGS. **1-3**, canopy **510** can include front surface **512** (e.g., a door), back surface **514**, side surface **516**, and top surface **518**. Canopy **510** can further include first section **522**, second section **524**, third section **526**, fourth section **530**, and fifth section **540**. As shown in FIG. **2**, first section **522** (e.g., leftmost side) can be disposed between base system **300** and canopy sleeve **560***a*, second section **524** (e.g., rightmost side) can be disposed between base system **300** and canopy sleeve **560***d*, third section **526** (e.g., center) can be disposed between canopy sleeves **560***b*, **560***c*, fourth section **530** (e.g., left side) can be disposed between canopy sleeves **560***a*, **560***b*, and fifth section **540** (e.g., right side) can be disposed between canopy sleeves **560***c*, **560***d*.
[0051] In some embodiments, canopy **510** can include a plurality of canopy sleeves **560**. For example, as shown in FIGS. **1-3**, canopy **510** can include four canopy sleeves **560***a*, **560***b*, **560***c*, **560***d*. Canopy sleeves **560** can be configured to secure frame system **400** to canopy system **500**. In some embodiments, canopy sleeves **560** and canopy **510** can be formed from the same material. For example, canopy sleeves **560** and canopy **510** formed from the same material can increase manufacturing efficiency and decrease cost. In some embodiments, for example, stitching and/or

fabric thickness can be adjusted to form canopy sleeves **560** and canopy **510** from the same material. In some embodiments, canopy sleeves **560** and canopy **510** can be integrally formed. In some embodiments, canopy sleeves **560** and canopy **510** can be formed from different materials. In some embodiments, canopy **510** and/or canopy sleeves **560** can include an elastic material.

Example Annex System

[0052] In some embodiments, tent system **200** can include an annex system **700** to expand an interior space of the tent system **200** and extend the interior space towards the ground. In some embodiments, annex system **700** couples to the base system **300** and/or canopy system **500**. As shown in FIGS. **1-4** and **16**, for example, annex system **700** can include annex floor **706**, annex canopy **708**, and/or annex frame assembly **710**. Base system **300** can include base bracket **702** configured to couple with annex frame assembly **710** as shown, for example, in FIGS. **4-7**. As will be described in further detail below, annex system **700** can couple with canopy system **500** to extend the weather protection (e.g., rain, sun, heat, wind, snow, cold, etc.) provided by the canopy system **500** to the ground.

[0053] In some embodiments, annex frame assembly **710** provides support to annex canopy **708** and couples to a base bracket **702** of base system **300** as shown, for example in FIGS. **1-7**. In some embodiments, annex frame assembly **710** is slideably coupled with base system **300** at locations on both sides of tent system **200**. Both annex frame assembly **710** and base system **300** include attachment features (e.g. base bracket **702** and fastening member **774**) on a right side and a left side of tent system **200**. Annex frame assembly **710** is configured so that a user can assemble one side of annex frame assembly **710** with one hand. For example, a user can slideably engage and couple a left side of annex frame assembly **710** with one hand, and slideably engage and couple a right side of annex frame assembly **710** with their other hand. Annex frame assembly **710** couples to an underside of base system **300**, for example, so that the annex frame assembly does not extend through canopy **510** of canopy system **500**.

[0054] In some embodiments, base bracket **702** is disposed on an underside of first base member **312** of base system **300**. In the illustrative embodiment shown in FIGS. **4** and **5**, base bracket **702** is fastened to first base member **312** with rivets, bolts, screws, or similar fastening devices. In some embodiments, base bracket **702** can be formed with first base member **312** such that base bracket **702** is integral with first base member **312**. In some embodiments, base bracket can be disposed on an underside of second base member **314** such that an annex system can be assembled to the opposite side of tent system **200**.

[0055] In some embodiments, base bracket **702** includes upper wall **720**, first side wall **722**, second side wall **724**, first flange **726**, and/or second flange **728**. In some embodiments the flanges **726**, **728** form an interior space **730** therebetween, as shown in FIGS. **5-7**. In some embodiments, upper wall **720** is coupled with the underside of first base member **312**, for example, at a distal end **313** of first base member **312**. In some embodiments, first side wall **722** extends from an edge of upper wall **720** and is perpendicular to upper wall **720**. In some embodiments, second side wall **724** is spaced apart from first side wall **722** and extends from an opposite edge of upper wall **720**. In some embodiments, second side wall **724** is perpendicular to upper wall **720**. In some embodiments, first flange **726** extends toward second side wall **724** from (e.g., a distal end of) first side wall **722**. Second flange **728** extends toward first side wall **722** from (e.g., a distal end of) second side wall **724**. In some embodiments, first flange **726** and second flange **728** are vertically aligned on the same plane and spaced apart to form a slot or gap **731** therebetween. In some embodiments, first flange **726** includes first surface **733** that faces upper wall **720** and second surface **735** opposite first surface **733** that faces away from base bracket **702**. In some embodiments, second flange **728** includes first surface **737** that faces upper wall **720** and second surface **739** opposite first surface **737** that faces away from base bracket **702**.

[0056] In some embodiments, interior space **730** is formed between upper wall **720**, first side wall **722**, second side wall **724**, first flange **726**, and second flange **728** as shown, for example, in FIG.

6. Interior space **730** is configured to receive fastening member **774** of annex frame assembly **710**. In some embodiments, base bracket **702** includes stopper **732**, for example extending from upper wall **720**, that extends into interior space **730** as shown in FIGS. **6** and **7**. Stopper **732** is configured to block movement of annex frame assembly **710** beyond a predetermined assembly distance in base bracket **702**. When the annex frame assembly **710** hits the stopper **732**, it provides feedback to the user that that annex frame assembly **710** is in the proper location to secure it to the base bracket **710**.

[0057] In some embodiments, annex frame assembly **710** includes one or more canopy support pole **770** to support annex canopy **708**, and fastening member **774** to couple with base bracket **702** as shown, for example, in FIGS. **5-7**. In some embodiments, canopy support pole **770** includes support portion **771** that provides support and shape to annex canopy **708**, and connection portions **772** that couple with base brackets **702** on either side of base system **300**. In some embodiments, connection portion **772** and support portion **771** can be separate components that are coupled together. In the illustrative embodiment in FIGS. **3** and **4**, connection portion **772** and support portion **771** are integrally formed such that canopy support pole **770** is a single component. Connection portion **772** is orientated horizontally relative to the ground and locates beneath base bracket **702**. In some embodiments, support portion **771** forms a C-shape and extends away from connection portion **772**, for example, at approximately 45 degrees relative to and away from the ground and then spans across the width annex system **700**.

[0058] In some embodiments, fastening member **774** couples with base bracket **702**. In some embodiments, fastening member **774** includes T-bolt **776** and actuator **778** as shown, for example in FIGS. **5-7**. In some embodiments, T-bolt **776** extends through connection portion **772** of canopy support pole **770** and couples with actuator **778**. In some embodiments, T-bolt **776** is slideably disposed in interior space **730** of base bracket **702**. In some embodiments, T-bolt **776** is supported by first surfaces **733**, **737** of first and second flanges **726**, **728**. In some embodiments, first and second flanges are located between and engage with T-bolt **776** and connection portion **772**. In the illustrative embodiment in FIG. **6**, connection portion **772** has a circular cross-section and second surfaces **735**, **739** have a concave shape to provide a greater surface area between the annex system **700** and the base system **300** to share load. In some embodiments, connection portion **772** and second surfaces **735**, **739** can have corresponding shapes to share loads therebetween that are non-circular. In the illustrative embodiment shown in FIGS. **6** and **7**, actuator **778** is a knob and threadably engaged with T-bolt **776** such that when actuator **778** is rotated, a distance between T-bolt **776** and connection portion **772** is reduced such that T-bolt **776** and connection portion **772** couple with first and second flanges **726**, **728**. In some embodiments, actuator **778** can be a lever, handle, or other user interface and configured to couple T-bolt **776** and connection portion **772** and base bracket **702** together.

[0059] By way of example, during assembly, a user can align T-bolt **776** of the annex frame assembly **710** with the interior space **730** of base bracket **702** and slide T-bolt **776** into interior space **730**. T-bolt **776** can slide along first surfaces **733**, **737** until T-bolt **776** engages with stopper **732**. Stopper **732** blocks further movement of annex frame assembly **710** so that T-bolt **776** does not extend beyond the end of base bracket **702**. Stopper **732** also provides sufficient contact length between connection portion **772** of canopy support pole **770** and base bracket **702** so that any moments acting on annex system **700** are accommodated. Stopper **732** can also be positioned to maintain desired dimensions of the annex system **700** relative to the canopy system **500**.

[0060] In some embodiments, annex canopy **708** couples to canopy system **500** so that annex canopy **708** surrounds front surface **512**, including the front door, of canopy **510** as shown, for example, in FIGS. **1-3**. In some embodiments, annex canopy **708** extends outwardly away from front surface **512** and is supported by canopy support pole **770** of annex frame assembly **710**. In some embodiments, annex canopy **708** is disposed around the canopy support pole **770** and extends towards the ground, for example, so that it locates adjacent to annex floor **706**. In some embodiments, annex canopy **708** has approximately vertical orientation as it extends towards the ground. In the illustrative embodiment shown in FIGS. **1-3**, annex canopy **708** tapers outwardly as it extends towards the ground such that a horizontal cross-sectional area of the annex canopy **708** is greater near the bottom than at the top. Annex canopy **708** provides weather protection (e.g., rain, sun, heat, wind, snow, cold, etc.) and can be made from similar materials to canopy **510** as described above. Annex canopy **708** extends the interior space of the tent system **200** to provide additional space away from the main tent sleeping area, for example, for storage, getting changed, keeping wet items out of the main tent interior space, or to provide an area for pets to rest.

[0061] In some embodiments, annex canopy **708** includes annex upper portion **712** and lower opening portion **716** as shown, for example, in FIGS. **1-3**. In some embodiments, annex upper portion **712** and lower opening portion **716** can be coupled together by stitching, gluing, or other suitable weather-proofing methods. In some embodiments, annex upper portion **712** and lower opening portion **716** can be formed from a continuous material.

[0062] In some embodiments, annex upper portion **712** couples with canopy **510** adjacent to canopy sleeve **560***d*, for example, between second section **524** and fifth section **540** as shown, for example, in FIGS. **1-3**. In some embodiments, annex upper portion **712** surrounds front surface **512**, and the front door of the tent, so that a user can remain protected from the weather as they exit the tent and enter the annex. In some embodiments, annex upper portion **712** is supported by and coupled to canopy support pole **770** of annex frame assembly **710**.

[0063] In some embodiments, annex upper portion **712** includes canopy connection portion **718**, first side section **734**, second side section **736**, top section **738**, front section **740**, and annex canopy sleeve **742** as shown, for example, in FIGS. **1-3**, **14**, and **15**. In some embodiments, first side **734** and second side **736** of annex upper portion **712** wrap around each side of second section **524** of canopy **510** to form side walls of the annex upper portion **712**. In some embodiments, top section **738** is disposed between first side **734** and second side **736**. In some embodiments, first side **734**, second side **736**, and top section **738** extend approximately parallel to fifth section **540** of canopy **510** and extend away from canopy **510** in a lateral direction to the side of vehicle **342**. In some embodiments, top section **738** extends approximately horizontal to canopy support pole **770** of annex frame assembly **710** and then transitions into front section **740** with approximately vertical orientation.

[0064] In some embodiments, canopy connection portion **718** extends across the interface of first side **734**, top section **738**, and second side **736** with canopy **510**, as shown, for example, in FIGS. **2**, **3**, **14**A, and **15**. In some embodiments, canopy connection portion **718** couples with canopy **510** along the interface between the section **524** and fifth section **540**. In some embodiments, canopy connection portion **718** includes a connection mechanism that couples with a corresponding connection mechanism on canopy **510**. For example, in some embodiments, canopy connection portion **718** can include a zipper that couples with a corresponding zipper on canopy **510**. In some embodiments canopy connection portion **718** can couple with canopy **510** using a hook and loop fastener, where the hook portion is on one area and the loop portion is on a corresponding area. In some embodiments, canopy connection portion **718** can include buttons or buckles that couple with corresponding holes formed in or loops attached to canopy **510**.

[0065] In some embodiments, canopy connection portion **718** can extend along a majority portion of the interface between annex system **700** and canopy system **500**. In the illustrative embodiment in FIGS. **14**A-C, canopy connection portion **718** includes first zipper **744** and second zipper **746**. In some embodiments, first zipper **744** and second zipper **746** are spaced apart at a central location above the door and front surface **512**. In some embodiments, each of the first zipper **744** and the second zipper **746** extend away from each other along the interface between annex system **700** and canopy system **500** and toward the hinge **318**. In the illustrative embodiment in FIG. **15**, canopy connection portion **718**′ includes first zipper **744**′, second zipper **746**′, and third zipper **748**′. In some embodiments, first zipper **744**′ spans the width of the door and front surface **512** along the interface between annex system **700** and canopy system **500**. In some embodiments, second zipper **746**′ is spaced apart from a first end **745** of first zipper **744**′ and extends away from first zipper **744**′ along the interface between annex system **700** and canopy system **500** and toward the hinge **318**. In some embodiments, third zipper **748**′ is spaced apart from a second end **747**, opposite the first end **745**, of first zipper **744**′ and extends away from first zipper **744**′ along the interface between annex system **700** and canopy system **500** and toward the hinge **318** on the opposite side of tent system **200** to the second zipper **746**′.

[0066] In some embodiments, canopy system **500** includes an annex flap **704** that covers and conceals canopy connection portion **718** as shown, for example, in FIG. **13**. Annex flap **704** provides weather protection (e.g., rain, sun, heat, wind, snow, cold, etc.) to the zippers **744**, **746** of canopy connection portion **718**. For example, annex flap **704** can include a sheet of material that extends along the interface between the annex system **700** and the canopy system **500** and positioned above the canopy connection portion **718** so that the annex flap **704** covers canopy connection portion **718**. Annex flap **704** can be made from similar materials to canopy **510** as described above.

[0067] In some embodiments, annex upper portion **712** can include annex canopy sleeve **742**. Annex canopy sleeve **742** can be configured to secure annex canopy **708** to annex frame assembly **710**. In some embodiments, annex canopy sleeve **742** can extend into an interior space of the annex system **700**. In some embodiments, annex canopy sleeve **742** can extend above top section **738** of annex upper portion **712**. In some embodiments, annex canopy sleeve **742** can include a sleeve recess (e.g., a pocket) configured to receive canopy support pole **770**. In some embodiments, annex canopy sleeve **742** can include a zipper sleeve to receive and enclose at least a portion of a canopy support pole **770**.

[0068] Annex canopy sleeve **742** can be formed from the same material and annex canopy **708** or canopy **510**. For example, forming annex canopy sleeve **742**, annex canopy **708**, and canopy **510** from the same material can increase manufacturing efficiency and decrease cost. In some embodiments, for example, annex canopy sleeve **742** can be coupled to annex canopy **708** by stitching or annex canopy sleeve **742** can be integrally formed with annex canopy **708**. In some embodiments, annex canopy sleeve **742** and annex canopy **708** can be formed from different materials.

[0069] In some embodiments, lower opening portion **716** provides access to the interior of the annex system **700** and tent system **200**. In some embodiments, lower opening portion **716** includes first side panel **780**, first door panel **782**, first corner panel **784**, front door panel **786**, second corner panel **788**, second door panel **790**, and/or second side panel **792** as shown in FIGS. **1-3**. In some embodiments, first side panel **780**, first door panel **782**, and first corner panel **784** extend from first side **734** of annex upper portion **712**. In some embodiments, first corner panel **784**, front door panel **786**, and second corner panel **788** extend from front section **740** of annex upper portion **712**. In some embodiments, second corner panel **788**, second door panel **790**, and second side panel **792** extend from second side **736** of annex upper portion **712**. In some embodiments, panels of the lower opening portion **716** are coupled with sections of the annex upper portion **712** by stitching, gluing or other similar coupling methods. In some embodiments, panels of the lower opening portion **716** are formed from the same material as sections of the annex upper portion **712**. In some embodiments, each of the panels **780**, **782**, **784**, **786**, **788**, **790**, **792** of lower opening portion **716** include anchor points that can be fixed to the ground to secure the annex system **700** to the ground and provide tension in the panels to create the desired shape of the annex canopy **708**.

[0070] In some embodiments, each of first door panel **782**, front door panel **786**, and second door panel **790** can be opened to gain access tent system **200**. For example, each of the door panels **782**, **786**, **790** can be rolled up and held in an open configuration, for example, by toggle and loop configurations included in the top and base of each panel. In the illustrative embodiment in FIG. **12**, each of the door panels **782**, **786**, **790** can include window meshes **791**, **793** so that air can pass through the annex system **700** when the panel doors **782**, **786**, **790** are in a closed configuration. In some embodiments, lower opening portion **716** includes third side panel **795** adjacent and parallel with vehicle **342**, and first, second, and third side panels **780**, **792**, **795** couple with the first base member **312** of base system **300**. In some embodiments, first, second, and/or third side panels **780**, **792**, **795** include guide rope **794** along portions of an upper perimeter of the side panels **780**, **792**, **795**. In some embodiments, base system **300** includes attachment clips **703**, for example, disposed on the underside of first base member **312** as shown, for example, in FIG. **4**. In some embodiments, the guide rope **794** portions couple with attachment clips **703** disposed on the underside of first base member **312** of base system **300** (example shown in FIG. **17**).

[0071] In some embodiments, annex floor **706** is disposed on the ground under annex canopy **708** and provides a weather-proof surface for the user to sit or stand on, or store items on. In the illustrative embodiment in FIG. **16**, annex floor **706** has square or rectangular shape that corresponds to the cross-sectional shape of annex canopy **708** at the interface with the ground. In some embodiments, annex floor **706** can be oversized and extend beyond the section panels of lower opening portion **716** to provide additional area at a campsite, for example, to sit on or place items on that is spaced apart from wet mud or similar. In some embodiments, annex floor **706** can include anchor points to anchor the annex floor **706** to the ground. In some embodiments, the anchor points in annex floor **706** can correspond to the anchor points in the lower opening portion **716** of annex canopy **708**.

Example Height Extension Portion of Annex Canopy

[0072] In some embodiments, annex canopy **708** includes height extension portion **714** as shown, for example, in FIGS. **8-12**. In some embodiments, height extension portion **714** is located between annex upper portion **712** and lower opening portion **716**. In some embodiments, height extension portion **714** can be coupled with annex upper portion **712** and lower opening portion **716** by stitching, gluing, or other suitable weather proofing methods. In some embodiments, each of annex upper portion **712**, height extension portion **714**, and lower opening portion **716** can be formed from a continuous material.

[0073] Height extension portion **714** adjusts a height of annex canopy **708** between an extended height H.sub.E (e.g., as shown in FIG. **8**), a shortened height H.sub.S (e.g., as shown in FIG. **9**), and/or a plurality of discrete heights between extended height H.sub.E and shortened height H.sub.S. Height extension portion **714** adjusts a height of annex canopy **708** so that tent system **200** can be used on both a tall vehicle, such as an SUV, and a low vehicle, such as a sedan or station wagon. For example, if tent system **200** is coupled to a low vehicle, the annex canopy **708** can be adjusted to a shorter height via height extension system **714** to avoid excess annex canopy material laying on the ground. In contrast, if the tent system **200** is coupled to a tall vehicle, the annex canopy **708** can be adjusted to a taller height using the height extension portion **714** so that the annex canopy **708** extends all the way to the ground.

[0074] In some embodiments, height extension portion **714** includes first extendable section **750**, second extendable section **752**, and/or third extendable section **754**, In the illustrative embodiment shown in FIG. **8**, first extendable section **750** extends below annex upper portion **712**, third extendable section **754** extends above lower opening portion **716**, and second extendable section **752** locates below first extendable section **750** and above third extendable section **754**. First extendable section **750** has height H**1**. Second extendable section **752** has height H**2**. Third extendable section **754** has height H**3**. In some embodiments, height H**1**, height H**2**, and height H**3** are the same. In some embodiments, height H**1**, height H**2**, and height H**3** are different.

[0075] In some embodiments, height extension portion **714** includes upper coupling member **756**, first lower coupling member **758**, second lower coupling member **760**, and/or third lower coupling member **762** as shown, for example, in FIGS. **8** and **10**. In the illustrative embodiment in FIG. **10**, the coupling members **756**, **758**, **760**, **762** are zippers and extend around the perimeter of height extendable portion **714**. In some embodiments, each of coupling members **756**, **758**, **760**, **762** can be toggle and hoops, button and holes, hook and loop, or any other suitable coupling configuration. In some embodiments, upper coupling member **756** is coupled to an upper edge of first extendable section **750**. In some embodiments, first lower coupling member **758** is coupled at the interface between a lower edge of first extendable section **750** and an upper edge of second extendable section **752**. In some embodiments, second lower coupling member **760** is coupled at the interface between a lower edge of second extendable section **752** and an upper edge of third extendable section **754**. In some embodiments, third lower coupling member **762** is coupled at a lower edge of third extendable section **754**.

[0076] The height of annex canopy **708** is extended height H.sub.E when none of the coupling members **756**, **758**, **760**, **762** are coupled to each other. The height of annex canopy **708** can be adjusted shorter than the extended height H.sub.E by coupling different coupling members **756**, **758**, **760**, **762** in different combinations. For example, the height of the annex canopy **708** can be adjust to shortened height H.sub.S by coupling third lower coupling member **762** with upper coupling member **756**. In the shortened height H.sub.S configuration, the lower edge of the third extendable section **754** couples with the upper edge of the first extendable section **750** and the height of the height extension portion **714** of the annex canopy **708** is removed from the total height of the annex canopy **708**.

[0077] In a first example configuration, illustratively shown in FIG. **10**, second lower coupling member **760** can be coupled with first lower coupling member **758** to reduce the height of annex canopy **708** by height H**2**. In this first example configuration, second extendable section **752** is folded over so that the upper and lower edges of second extendable section **752** become connected.

[0078] In a second example configuration, third lower coupling member **762** can be coupled with first lower coupling member **758** to reduce the height of annex canopy **708** by the combined height of height H**2** plus height H**3**. In this second example configuration, second extendable section **752** and third extendable section **754** are folded over so that the upper edge of second extendable section **752** is coupled to the lower edge of third extendable section **754**.

[0079] In an illustrative example shown in FIGS. **11** and **12**, second lower coupling member **760** couples with upper coupling member **756**. In the illustrative configuration, the height of annex canopy **708** has been shortened by the combined height of height H**1** plus height H**2**. In the illustrative configuration, only the third extendable section **754** is visible from the interior of the annex system **700**. In the illustrative configuration, first and second extendable sections **750**, **752** are folded over on the outside of annex system **700**.

[0080] In some embodiments, height extension portion **714** can include more than three extendable portions to provide additional adjustment in the height of the annex canopy **708**. In some embodiments, height extension portion **714** can include less than three extendable portions to provide additional adjustment in the height of the annex canopy **708**. In some embodiments, the height extension portion can adjust an annex canopy from a height of about 2300 mm to about 1500 mm. In some embodiments, the height extension portion can adjust an annex canopy from a height of about 2150 mm to about 1700 mm. In some embodiments, the height extension portion can adjust an annex canopy from a height of about 2100 mm to about 1850 mm. In some embodiments, the height extension portion can adjust an annex canopy from a height of about 2050 mm to about 1900 mm. In some embodiments, the height extension portion can adjust an annex canopy from a height of about 2050 mm to about 1950 mm.

[0081] It is to be appreciated that the Detailed Description section, and not the Brief Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections may set forth one or more but not all embodiments of the support assembly system and apparatus, and thus, are not intended to limit the present embodiments and the appended claims.

[0082] The present disclosure has been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the

description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.
[0083] The foregoing description of the specific embodiments will so fully reveal the general nature of the disclosure that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the present disclosure. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the teachings and guidance.
[0084] The breadth and scope of the present disclosure should not be limited by any of the above-described embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A tent system, comprising: a base configured to mount to a vehicle; a base canopy; a frame system coupled to the base and configured to support the base canopy; and an annex comprising an annex canopy and an annex frame assembly configured to support the annex canopy, wherein the annex is removably coupled to the base and the base canopy.
2. The tent system of claim 1, wherein the annex canopy is removably coupled to the base canopy and the annex frame assembly is removably coupled to the base.
3. The tent system of claim 1, wherein the annex canopy is removably coupled to the base canopy with at least one zipper.
4. The tent system of claim 1, wherein the annex frame assembly is coupled to a bottom side of the base.
5. The tent system of claim 1, wherein the base canopy comprises a door disposed on a first surface, and the annex canopy surrounds the door.
6. The tent system of claim 1, further comprising a ladder coupled to the base and extending between the floor and the base, wherein the annex completely surrounds the ladder.
7. The tent system of claim 1, wherein the annex canopy includes a lower opening portion, the lower opening portion having a plurality of side sections and at least two of the plurality of side sections are configured to be in one of an opened configuration and a closed configuration.
8. A tent system, comprising: a base configured to mount to a vehicle; a base canopy; a frame system coupled to the base and configured to support the base canopy; and, an annex comprising an annex canopy and an annex frame assembly configured to support the annex canopy, wherein the annex frame assembly is removably coupled to a bottom side of the base.
9. The tent system of claim 8, wherein the annex frame assembly is removably coupled to a base bracket disposed on the bottom side of the base.
10. The tent system of claim 9, wherein the annex frame assembly comprises a fastening member configured to slideably engage with an interior space formed in the base bracket.
11. The tent system of claim 10, wherein the fastening member comprises a T-bolt slideably disposed in the interior space and an actuator threadably coupled with the T-bolt and configured to secure the annex frame assembly to the base bracket.
12. The tent system of claim 9, wherein the base bracket includes a first flange and a second flange spaced apart from the first flange to form a slot therebetween, the first and second flanges comprise engagement surfaces that have a shape corresponding to a connection portion of the annex frame assembly and are configured to engage the connection portion of the annex frame assembly.
13. The tent system of claim 12, wherein the connection portion of the annex frame assembly is

cylindrical and the engagement surfaces of the first and second flanges are concave.
14. The tent system of claim 10, wherein the base bracket includes a stopper disposed in the interior space and configured to block movement of the fastening member.
15. The tent system of claim 8, wherein the annex frame assembly does not extend through the base canopy.
16. A tent system, comprising: a base configured to mount to a vehicle; a base canopy; a frame system coupled to the base and configured to support the base canopy; and, an annex configured to extend an interior space of the tent system in a lateral direction away from the base canopy and in a vertical direction toward the ground, the annex comprising an annex canopy and an annex frame assembly configured to support the annex canopy, wherein the annex canopy includes a height extendable portion configured to adjust a height of the annex canopy relative to the ground.
17. The tent system of claim 16, wherein the height extendable portion comprises a first extendable section having a first section height, an upper coupling member disposed on an upper edge of the first extendable section, and a first lower coupling member disposed on a lower edge of the first extendable section, wherein the annex canopy is configured to be disposed in a first shortened configuration such that the first lower coupling member is coupled to the upper coupling member and the height of the annex canopy is reduced by the first section height.
18. The tent system of claim 17, wherein the upper coupling member and the lower coupling member comprises a zipper, a toggle and loop, a button and hole, or a hook-and-loop fastener.
19. The tent system of claim 17, wherein the upper coupling member extends along a majority of the upper edge of the first extendable section, and the first lower coupling member extends along a majority of the lower edge of the first extendable section.
20. The tent system of claim 17, wherein the height extendable portion further comprises a second extendable section having a second section height and extending from the lower edge of the first extendable section, the first lower coupling member disposed on an upper edge of the second extendable section, and a second lower coupling member disposed on a lower edge of the second extendable section.
21-26. (canceled)
